

ZYVORAI LABS · CUSTOMER DOCUMENTATION

cloud-netconfig

Page-by-Page Product Manual

2026-08-29

Page-by-page guides

Each guide follows: Purpose → When to use it → How to get there → What you can do → Related pages.

Every route is also listed in the [complete page index](#).

Admin

Page	What it covers
Admin Basics	Packaging, systemd unit, local HTTP API, privileges, and ops.

Onboarding

Page	What it covers
Getting Started	Install cloud-netconfigd and verify metadata-driven networking on a cloud VM.

Operations

Page	What it covers
Workflows	Day-2 validate, reload, multi-interface rollout, and upgrade jobs with cnctl.

Setup

Page	What it covers
Configuration	YAML daemon config: logging, metadata refresh, supplementary interfaces, and routing.

4 guides. Regenerate: `node scripts/customer-docs/generate-guide-index.mjs` .

Admin Basics

Purpose

Packaging, systemd unit, local HTTP API, privileges, and ops.

How to get there

- Topic id: `admin-basics`
- Section: **Admin** → **Admin Basics**

Guide

Topic	Detail
Unit	<code>cloud-netconfigd.service</code> — notify type, watchdog, hardened syscall filter
Capabilities	<code>CAP_NET_ADMIN</code> only after privilege drop
API	<code>/health</code> , <code>/api/status</code> , <code>/api/cloud/status</code> on <code>server.listen</code>
State	<code>/run/cloud-network</code> — metadata persistence when <code>state.persist_metadata: true</code>
Completions	bash, zsh, fish installed under <code>/usr/share/</code>

Logs and service control

```
sudo systemctl status cloud-netconfigd
sudo journalctl -u cloud-netconfigd -r
```

Security notes

- Keep the HTTP listener on localhost unless you add TLS and access controls.
- Restrict `/etc/cloud-network/*.yaml` to root-readable permissions.
- Review `SECURITY.md` before exposing metadata-derived routes on shared tenants.

Operate (CLI)

1. Monitor via systemd and journald — there is no remote metrics endpoint in Community Edition.
2. After package upgrades, verify `systemctl cat cloud-netconfigd` matches shipped unit file.
3. **Empty / fail:** Check `cloud-network` user exists and `/run/cloud-network` is writable.
4. **Success:** Watchdog stays healthy; API returns `running` status.

Related pages

- [Getting Started](#)
- [Page index](#)

Getting Started

Purpose

Install cloud-netconfigd and verify metadata-driven networking on a cloud VM.

How to get there

- Topic id: `getting-started`
- Section: **Onboarding** → **Getting Started**

Guide

cloud-netconfig runs as a systemd daemon on Azure, AWS EC2, or GCP VMs. It watches instance metadata and netlink events, then configures supplementary interfaces, custom routing tables, and policy rules so multi-homed hosts route correctly.

Prerequisites

- A cloud VM with secondary network interfaces attached in the provider console.
- Rust toolchain and `make` for source builds, or a packaged release from GitHub.
- `sudo/root` for install and route management.

Install and enable

```
git clone https://github.com/hypersdk/cloud-netconfig.git
cd cloud-netconfig
make build && sudo make install
sudo cp /etc/cloud-network/config.yaml /etc/cloud-network/cloud-network.yaml
sudo systemctl enable --now cloud-netconfigd
```

First verification

```
cnctl status all
curl -s http://127.0.0.1:5209/api/status | jq
```

Troubleshooting

- **Service fails on bare metal or non-cloud VMs** — auto-detection requires provider metadata; use a supported cloud instance for testing.
- **Config not loaded** — ensure `cloud-network.yaml` exists (see [Configuration](#)).

Next steps

- [Configuration](#)
- [Common workflows](#)
- [Admin basics](#)

Operate (CLI)

1. After install, run `cnctl status all` before changing production routing.

2. Confirm `/api/status` returns `"status":"running"` and the expected provider string.
3. **Empty / fail:** Read `journalctl -u cloud-netconfigd` for metadata or netlink errors.
4. **Success:** Secondary NICs listed in config show addresses and policy routes in `ip rule` / `ip route show table` .

Related pages

- [Getting Started](#)
 - [Page index](#)
-

Workflows

Purpose

Day-2 validate, reload, multi-interface rollout, and upgrade jobs with `cnctl`.

How to get there

- Topic id: `workflows`
- Section: **Operations** → **Workflows**

Guide

Job	Commands
Health check	<code>cnctl status all</code> · <code>curl http://127.0.0.1:5209/health</code>
Config rollout	<code>cnctl apply --dry-run</code> → copy YAML → <code>systemctl restart cloud-netconfigd</code>
Reload	<code>sudo systemctl reload cloud-netconfigd</code> (after <code>cnctl reload</code> reminder)
Debug	Set <code>logging.level: debug</code> → restart → <code>journalctl -u cloud-netconfigd -f</code>
Upgrade	Stop service → replace binaries → start → <code>cnctl version</code>

Multi-interface rollout

1. Attach secondary NICs in the cloud console.
2. Copy `examples/multi-interface.yaml` and set `network.interfaces.enabled` .
3. Validate with dry-run, restart daemon, confirm routes with `ip rule` and `ip route show table 9999` (or your `table_base`).

Operate (CLI)

1. Prefer dry-run and journal inspection before restarting on production multi-homed hosts.
2. Use `cnctl status network` to confirm netlink state matches provider metadata.
3. **Empty / fail:** Capture logs and metadata connectivity (IMDS/GCP metadata endpoints).
4. **Success:** Traffic egresses via the intended interface per policy rules.

Related pages

- [Getting Started](#)
 - [Page index](#)
-

Configuration

Purpose

YAML daemon config: logging, metadata refresh, supplementary interfaces, and routing.

How to get there

- Topic id: `configuration`
- Section: **Setup → Configuration**

Guide

All daemon behaviour is driven by `/etc/cloud-network/cloud-network.yaml`. Key knobs:

- `network.interfaces.enabled` — supplementary NICs to configure from metadata.
- `network.routing` — custom table base (`table_base`), policy routing, default route management.
- `metadata.refresh_interval` — how often metadata is re-fetched (default `300s`).
- `cloud.*` — auto-detect or pin provider; tune Azure API version, AWS IMDSv1/v2, GCP recursive fetch.

Copy a provider example from `distribution/etc/cloud-network/examples/` as a starting point.

Validate

```
cnctl apply --dry-run
cnctl apply --dry-run -c distribution/etc/cloud-network/examples/aws.yaml
```

Apply

```
sudo cp my.yaml /etc/cloud-network/cloud-network.yaml
sudo systemctl restart cloud-netconfd
```

Troubleshooting

- **MTU override ignored** — the schema field is `mtu.override_value` (see annotated `config.yaml`).
- **Daemon ignores edits** — wrong filename; active path is `cloud-network.yaml`, not only the packaged `config.yaml`.

Next steps

- [Getting started](#)
- [Common workflows](#)
- [Admin basics](#)

Operate (CLI)

1. Stage changes in a non-production path; always `--dry-run` first.
2. Diff against examples for your cloud (`aws.yaml`, `azure.yaml`, `gcp.yaml`).
3. **Empty / fail:** Dry-run stderr shows YAML parse or duration validation errors.
4. **Success:** Restart completes; `cnctl status network` reflects new interfaces.

Related pages

- [Getting Started](#)
 - [Page index](#)
-

cloud-netconfig — Complete page index

Every primary navigable dashboard route.

Generated: 2026-08-29 · 4 routes

Regenerate: `node scripts/customer-docs/generate-page-index.mjs`

Onboarding

Page	Route	Purpose	Guide
Getting Started	<code>getting-started</code>	Install cloud-netconfigd and verify metadata-driven networking on a cloud VM.	Open

Setup

Page	Route	Purpose	Guide
Configuration	<code>configuration</code>	YAML daemon config: logging, metadata refresh, supplementary interfaces, and routing.	Open

Operations

Page	Route	Purpose	Guide
Workflows	<code>workflows</code>	Day-2 validate, reload, multi-interface rollout, and upgrade jobs with cnctl.	Open

Admin

Page	Route	Purpose	Guide
Admin Basics	<code>admin-basics</code>	Packaging, systemd unit, local HTTP API, privileges, and ops.	Open

Related

- [Customer docs home](#)
- [Page-by-page guides](#)